📌 EXPECTED OUTCOME (FINAL REPORT SECTION)

🎯 Project Outcome

By completing this project, the following outcomes were achieved:

◈ 1. Complete Data Preprocessing Workflow

A full end-to-end data preprocessing pipeline was successfully implemented, including:

- Data acquisition from multiple sources (CSV, JSON, SQL, API)
- Data cleaning and preparation
- Feature engineering and transformation

👉 This demonstrates the ability to handle real-world datasets.

◈ 2. Detailed Data Cleaning

The dataset was thoroughly cleaned using:

- Missing value imputation (mean, mode, KNN, MICE)
- Outlier detection and treatment (Z-score, IQR, Winsorization)

👉 Result:
A consistent and reliable dataset with minimal noise.

◈ 3. Advanced Encoding & Scaling

Different encoding techniques were applied:

- Ordinal Encoding
- Label Encoding
- One-Hot Encoding

Scaling methods used:

- StandardScaler
- MinMaxScaler
- RobustScaler

👉 Result:
Data is now suitable for machine learning algorithms.

◈ 4. Feature Engineering & Transformation

New meaningful features were created:

- Debt-to-Income Ratio
- Average Monthly Transactions
- Spending-to-Income Ratio

Transformations applied:

- Log, square root, and power transformations

👉 Result:
Improved predictive power of the dataset.

◈ 5. Machine Learning Readiness

The final dataset:

- Contains no missing values
- Has reduced outliers
- Is properly encoded and scaled
- Includes engineered features

👉 Therefore, it is **fully ready for building machine learning models**.

◈ 6. High-Quality Dataset Generated

A clean and structured dataset (final_cleaned_dataset.csv) was produced, which can be directly used for:

- Classification models
- Credit risk prediction

▯ FINAL LINE (VERY IMPORTANT)

“This project successfully demonstrates a complete data preprocessing and feature engineering pipeline, transforming raw data into a high-quality, machine learning-ready dataset.”